卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0404 30-B003701S-T0404

第 1 页 共 1 页
PAGE 1 OF 1

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

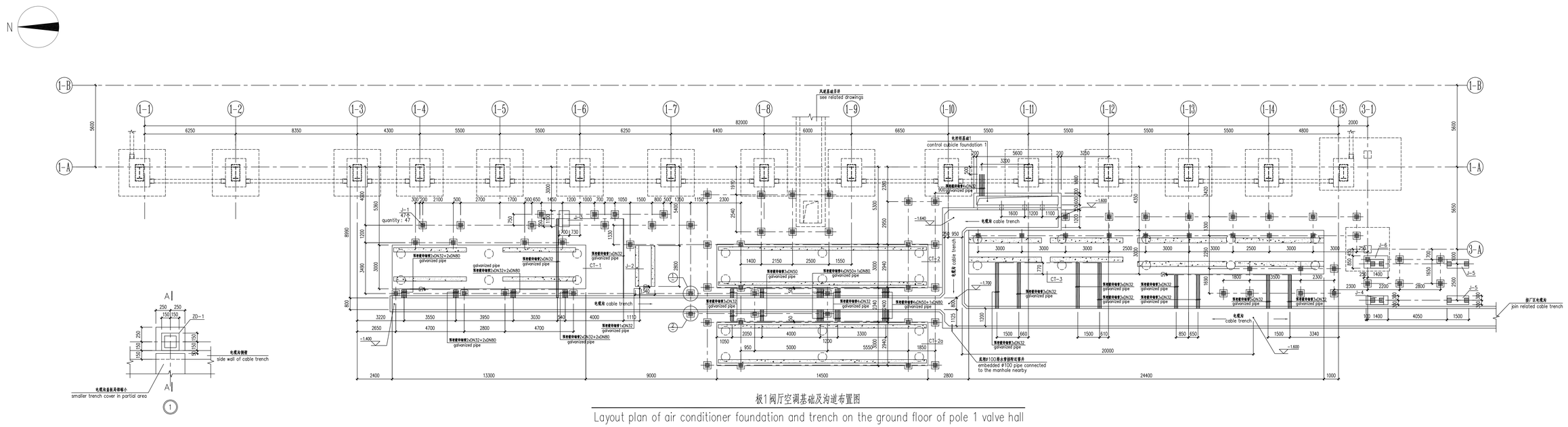
结构 专业 第 04 卷 第 04 册
Structure SPECIALITY VOLUME BOOK

卷册名称 阀厅空调基础图
Volume name Air conditioner foundation of valve hall

批 准 APPROVED BY	2018.01.05
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	18.01.05
图 纸 3 张 DRAWING NOS.	打印件 本 PRINTED COPIES NOS.

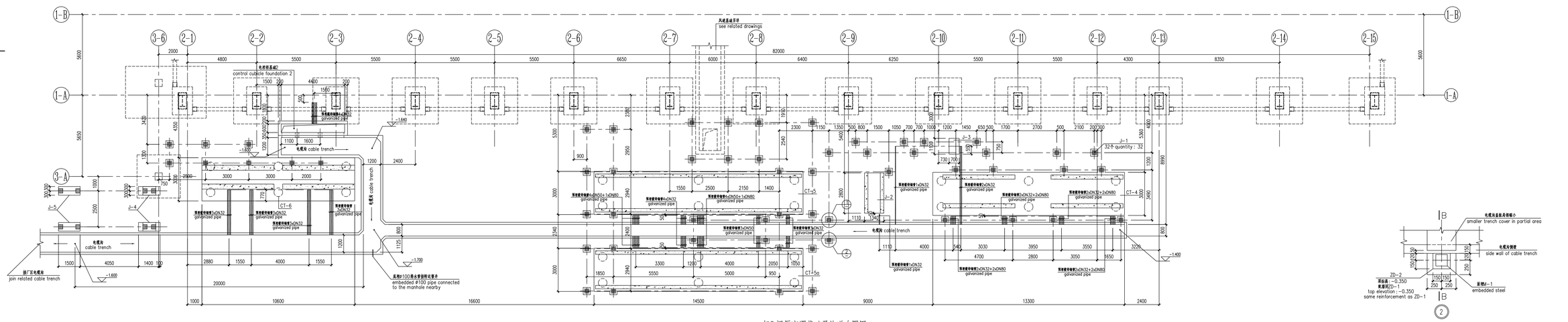
序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0404-001	阀厅空调基础及沟道布置图 Layout plan of air conditioner foundation and trench on the ground floor	A	1		
2	30-B003701S-T0404-002	设备基础详图一 Detail drawing 1	A	1		
3	30-B003701S-T0404-003	设备基础详图二 Detail drawing 2	A	1		

注:



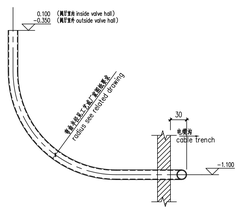
极1阀厅空调基础及沟道布置图

Layout plan of air conditioner foundation and trench on the ground floor of pole 1 valve hall



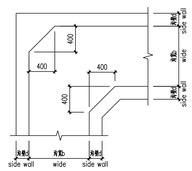
极2阀厅空调基础及沟道布置图

Layout plan of air conditioner foundation and trench on the ground floor of pole 2 valve hall



预埋镀锌钢管详图

Detail of galvanized pipe



电缆沟倒角详图

Detail of cable trench corner

附注：

1. 室内±0.000相当于绝对标高193.90(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。
2. 如无特别说明，本工程图纸所注尺的长度单位为毫米(mm)，标高单位为米(m)。
3. 混凝土
   基础：fc'=28MPa，垫层：fc'=21MPa。
4. 钢筋
   Φ表示 Grade40(fy=280MPa)钢筋，Φ表示 Grade60(fy=420MPa)钢筋，钢筋应符合标准 ASTM A615 or ASTM A706的规定。
5. 钢材
   钢板、角钢和扁钢采用Q235B。
6. 除注明外，≤No.19钢筋锚固长度不小于38d，≥No.22钢筋锚固长度不小于47d，d为钢筋直径。
7. 桩承台底面钢筋混凝土保护层厚度为100mm，其他部位基础混凝土保护层厚度：≤No.16钢筋为40mm；≥No.19钢筋为51mm。
8. 地基承载力特征不小于70kPa，超挖部分采用级配碎石回填，分层压实，压实系数≥0.94。
9. 所有外露铁件均采用热镀锌防腐。
10. 电缆埋管图中仅示意，埋管规格、数量、定位等要求均以相关专业或厂家图纸为准。
11. 电缆沟及盖板为钢筋混凝土结构，电缆沟排水坡度~0.5%，沟道每隔30米设变形缝一道，做法详见沟道详图。
12. A-A、B-B剖面见-02图。
13. 预埋件均以中心线定位。

Specification :

1. The indoor elevation of ±0.000 is equal to the absolute elevation of 193.90 (geodestic datum system EGM2008). If not specified,the elevations of the project are adopt to the relative elevation.
2. Unless otherwise specified, the length of the drawings is in millimeters (mm) and the elevation is in meters (m).
3. Concrete
   Foundations : fc'=28MPa , Cushions : fc'=21MPa.
4. Steel bars
   Φ represents the steel reinforcement of Grade40(fy=280MPa) and Φ represents Grade60(fy=420MPa). Steel bars should conform to ASTM A615 or ASTM A706.
5. Steel
   The structural carbon steel , grade Q235B, should be used for steel plate, angle steel and steel strip.
6. Unspecified anchorage length of rebars should no less than 38d for rebars ≤No.19, or 47d for rebars ≥No.22, d is the diameter of anchored rebars.
7. The thickness of concrete cover for reinforcements in bottom of the pile cap is 100mm, The thickness of concrete cover for reinforcements in other foundation is 40mm for rebars ≤No.16, and 51mm for rebars ≥No.19.
8. Ground bearing pressure ≥70kPa,the gap between blinding layer and natural floor should be refilled with graded gravel and tamped by layers,the compacting factor should be more than 0.94.
9. All structural steel should be galvanized.
10. Size ,quantity ,location of cable duct should comply with electrical engineering drawings related.
11. Cable trench and trench cover are reinforce concrete structure, the longitute slope for drainage at the bottom is i=0.5%, an expansion joint should be set at each 30m long of cable trench, see related drawings for detail drawings.
12. Section drawing A-A , B-B see drawing -02.
13. All the embedded steel are located by center axis.

电 气	郑煜	2018.01.03
水 布	沙鸥	2018.01.03
总 图	张吉祥	2018.01.03
暖 通	孔凡	2018.01.03
专 业	会 签	日 期
Speciality	Sign	Date

Reference Drawings

Drwg. No.	Title
30-B003701S-T0202-002	Plan layout of foundations for the control building
30-B003701S-T0801-002	Plan layout of pile cap for pole 1 valve hall and firewall
30-B003701S-T0806-002	Plan layout of pile cap for pole 2 valve hall and firewall

Revisions				
A	宫思懿 Name		2018.01.05 Date	First issue Notes

巴基斯坦联络至里一拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA EQUIPMENT & TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H Y D C - L C S - E C I - D D - C S - 0 4 0 4 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	宫思懿	2017.12.30
校核 Checked	宫思懿	2017.12.31
审核 Reviewed	刘明钢	2018.01.03
批准 Approved	王	2018.01.05

比例 Scale

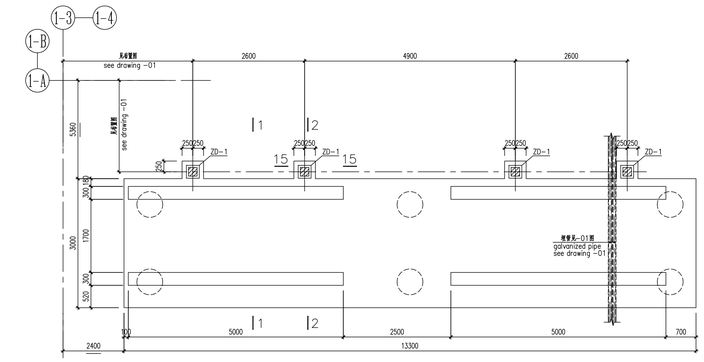
图号 DWG NO. 30-B003701S-T0404-001

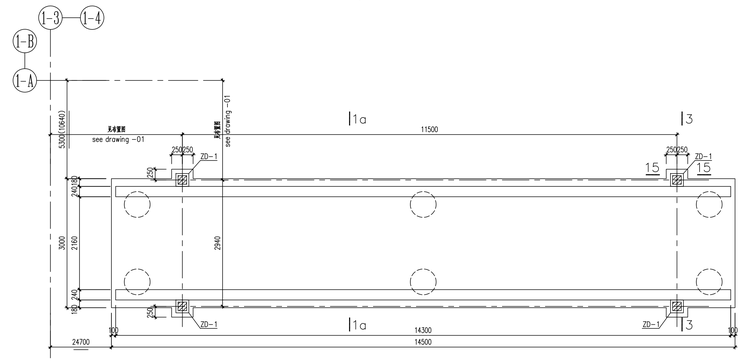
版本 REV. A

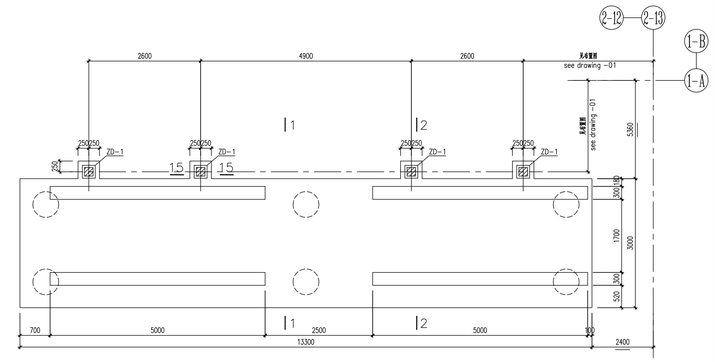
阀厅空调基础及沟道布置图

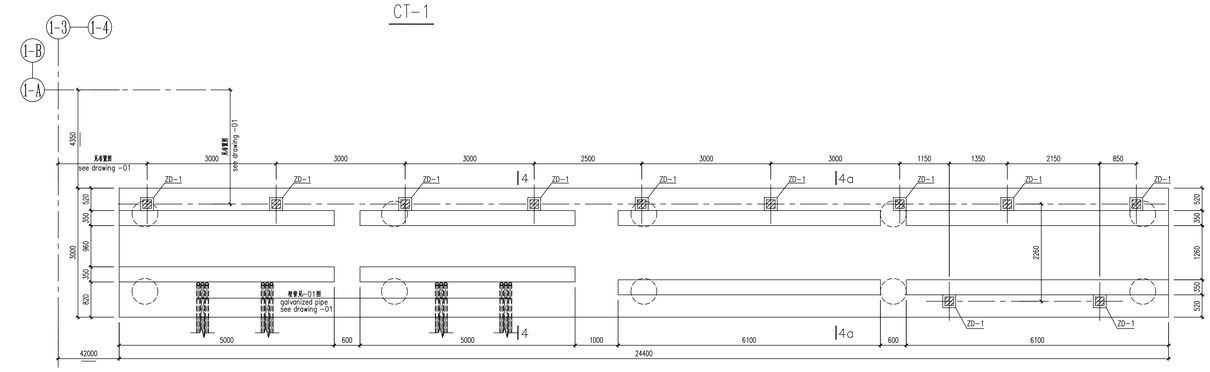
Layout plan of air conditioner foundation and trench on the ground floor

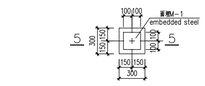
841X1189

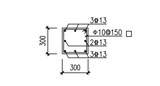


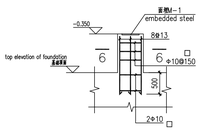


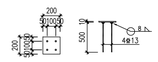


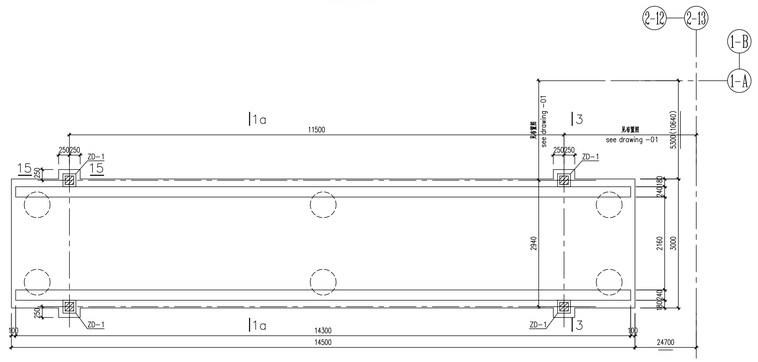


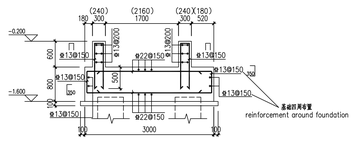


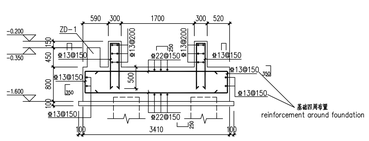


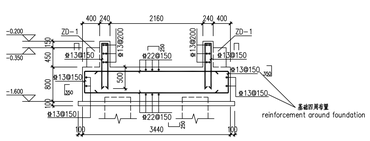


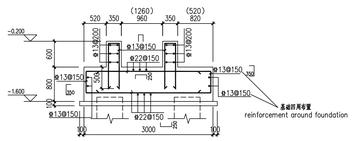


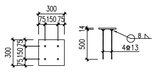


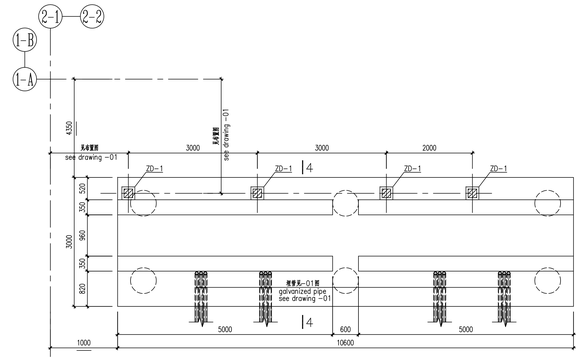


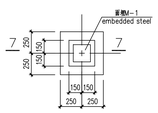


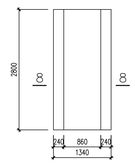


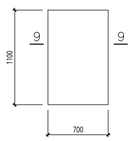


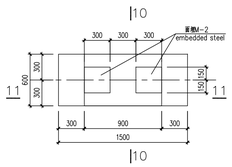


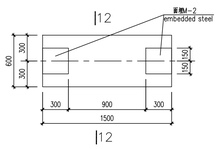


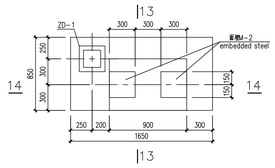


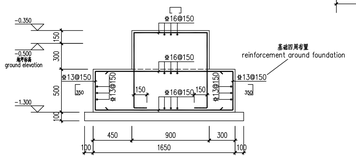


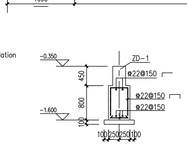


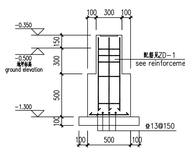


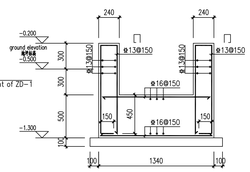


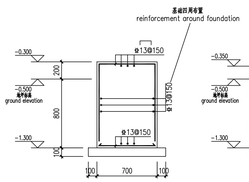


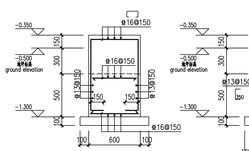


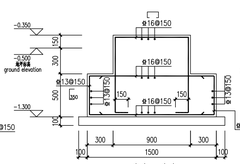


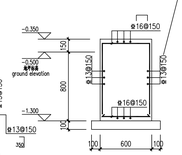


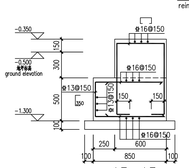


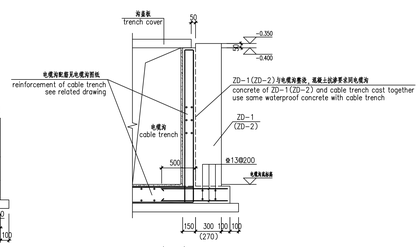












Reference Drawings	
Dwg. No.	Title

Revisions

A	首版图	2017.12.30	First issue
	Name	Date	Notes

巴基斯坦凯蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 4 0 4 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed		2017.12.30
校核 Checked		2017.12.31
审核 Reviewed		2018.01.03
批准 Approved		2018.01.05

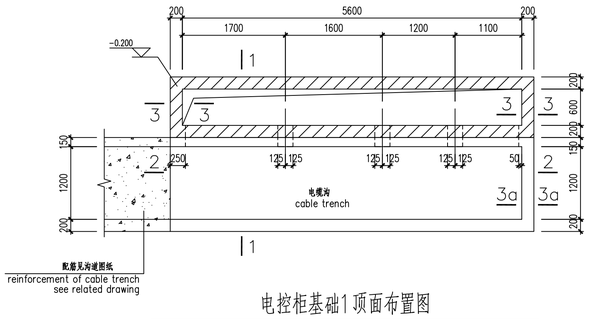
专业 Speciality	会签 Sign	日期 Date
暖通		2018.01.05

设备基础详图一

Detail drawing 1

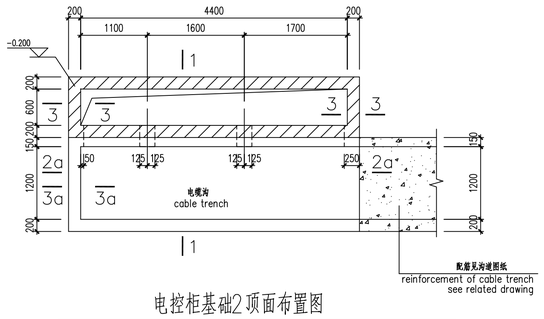
比例 Scale | 图号 DWG. NO. 30-B003701S-T0404-002 | 版本 REV. A

841X1189



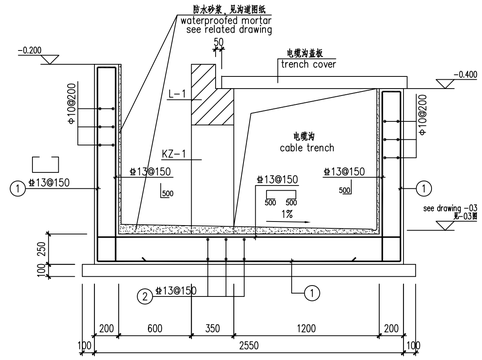
电控柜基础1顶面布置图

top floor of control cubicle foundation 1

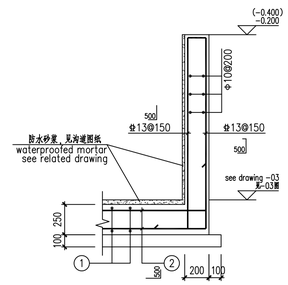


电控柜基础2顶面布置图

top floor of control cubicle foundation 2

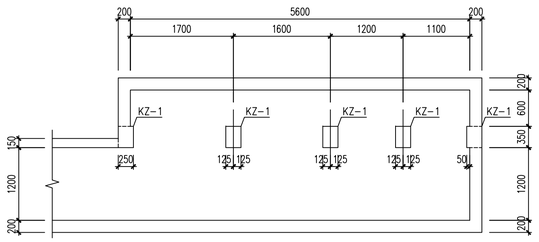


1-1



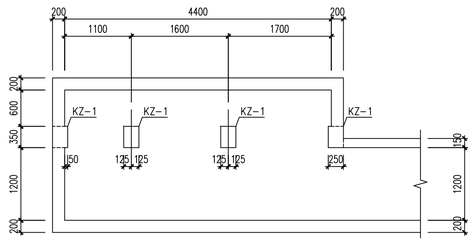
3-3

(3a-3a)



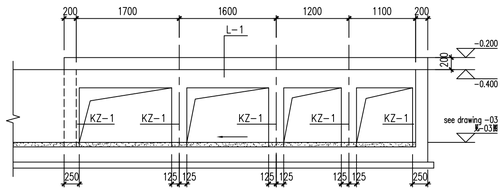
电控柜基础1底板布置图

ground floor of control cubicle foundation 1

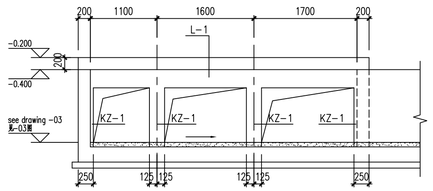


电控柜基础2底板布置图

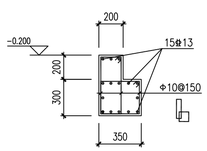
ground floor of control cubicle foundation 2



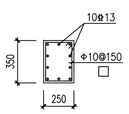
2-2



2a-2a



L-1



KZ-1

顶标高：-0.400

top elevation

附注：

1. 混凝土抗渗要求同电缆沟。

Specification :

1. Same requirements for underground concrete as cable trench.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes
A	官恩懿	2017.12.30	First issue

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-LCS-ECI-DD-CS-0404-0003 A

	姓名 Name	日期 Date
设计 Designed	官恩懿	2017.12.30
校核 Checked	李洪涛	2017.12.31
审核 Reviewed	孙刚	2018.01.03
批准 Approved	谢俊	2018.01.05
比例 Scale		图号 DWG NO.

设备基础详图二

Detail drawing 2

30-B003701S-T0404-003

版本 REV. A

专业 Speciality	会签 Sign	日期 Date
暖通	石成磊	2018.01.05

594X841